

Assignment 1

This project presents a photogrammetric reconstruction of Sculpture of Luke Kelly, acquired and processed using a Structure-from-Motion (SfM) and Multi-View Stereo (MVS) workflow. The sculptor was chosen because of its feature of curly hair which would prove difficult to reproduce, thus a challenge.

Images&Software

150 images captured using Pixel 9 Pro at [3072 x 4080 around 12MP]. Image overlap maintained 70–80% between adjacent views. Although the sculptor was too tall to be captured for more top-down view. Reality Scan was used on a 3050ti laptop.

Sparse Reconstruction

150/150 images were successfully aligned. Possibly due to rich and careful dataset and max feature per mpx and image doubled from default (20000 and 80000 respectively). Vertices [572,707] Faces[0]

Dense construction and Mesh reconstruction

Dense point cloud - [3 hours] of computation. At normal detail preset Vertices: [13,118,316], Faces: [26,118,316].Reconstruction region was tightened using high detail preset ,smoothing at 1.5x . UV unwrapping and diffuse texture baking were performed at [max - 8192*8192 px ,min-400*400 px] with Laplacian smoothing at 1.5 passes. Vertices: [2,499,828], Faces: [5,000,004].Great reduction but not enough.

Mesh Processing and Optimization

Post-processing in Blender - removal of artifacts; careful deletion of vertices of the top head; sculpting to close open area to not cause hollow lighting during Unity. Top head has unusual green color because the vertex were uv mapped thus had information baked into it.Decimation to approximately [150k faces].

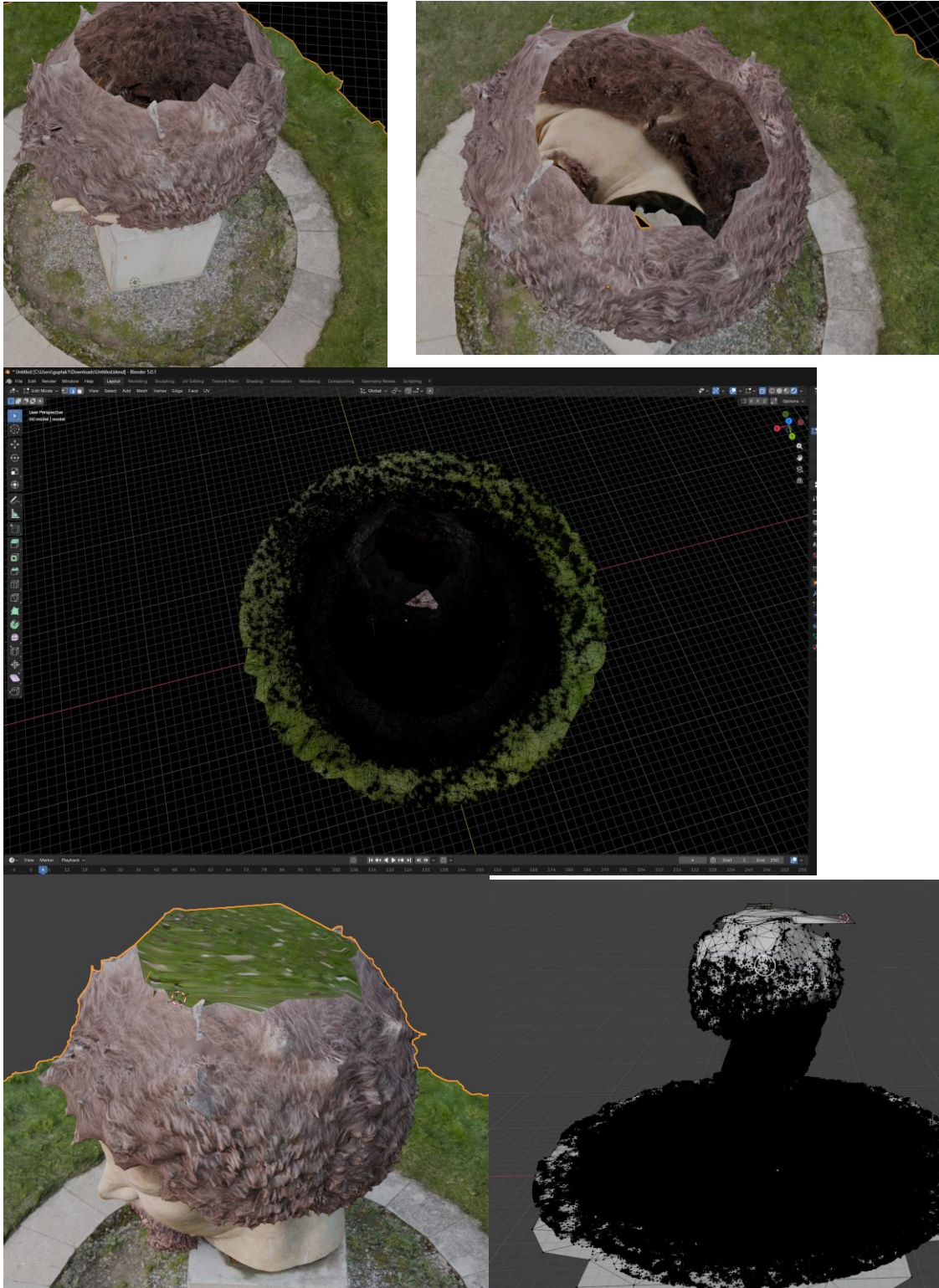
Evaluation and Limitations

Strengths include strong geometric fidelity and successful AR deployment. Limitations include partial reconstruction gaps due to restricted capture angles and computational intensity during dense reconstruction due to extreme number of images required to get a workable construction of hair features. Future improvements include extended vertical capture and improved oblique angle coverage and a better machine (my laptop BSOD'ed 3 times. Lg37 lab machines restarted thrice midway construction).



Initial construction

Kartik Gupta



Some midway processes

Kartik Gupta



Final Output